Mission MBBS TEAM

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HEC USAT Pre Test

COMPUTER SCIENCE

INSTRUCTIONS;

Duration of Test is 120 minutes (100 Questions).
No Negative Marking
Fill only one circle each question on given respective sheet .
Answer key will be displayed on your page Mission MBBS (Facebook) for MCQS discussion YouTube (Mission MBBS) .
Incase of any Confusion get help from Mission MBBS TEAM.

COMPUTER SCIENCE

1. Which component is responsible for processing data in a computer?
(a) Motherboard (b) CPU (c) RAM (d) Hard drive

2. What is the purpose of a router?
(a)To connect multiple computers to a network
(b) To control the flow of data on a network
(c)To store data on a network
(d) To provide internet access

3. Which programming language is often used for web development?
(a) C++ (b) Python (c) Java (d)Assembly

4. Which data structure is best suited for implementing a stack?
(a) Linked list (b) Array (c) Queue (d) Tree

5. Which SQL statement is used to retrieve data from a table?
(a) SELECT (b) INSERT (c) UPDATE (d) DELETE

6. What is the purpose of the <img> tag in HTML?
(a) To insert text (b) To create a link
(c) To insert an image (d) To define a heading

7. What is the main function of an operating system?

(a) To manage hardware and software resources (b)To run applications
(c) To store data (d)To provide internet access

8.Which programming language is known for its functional programming paradigm? (a) C++ (b) Python (c)Java (d) Haskell

9.What is artificial intelligence (AI)?
(a) The ability of machines to think and learn like humans
(b) The use of computers to solve complex problems
(c) The creation of virtual worlds
(d) The development of robots

10. RAM stand for .
(a) Random access memory (b) Random artificial memory
(c) Read access memory (d) None

11. ROM Stand for .
(a) Read only memory (b) Random only memory
(c) Read operate memory (d) none

12. The art and science of getting pages to rank higher in search engine such as google called ?
(a) CEO (b) SEO (C) both (d) none

13. The software which is used for documentation purpose such as text writing ID Card, Admission form, and office work .
(a) Ms Word (b) e book (c) PPT (d) none

14. Key board Shortcut used for paste in computer is ?
(a) ctrl+v (b) ctrl+ p (c) ctrl + c (d) ctrl + y

15. A program which that can be used for presentation also for create beautiful slideshow with text , audio, video is called .
(a) MS WORD (b) Power point (c) video editor (d) none

16.Which protocol is used for transferring files over the internet?
(a) HTTP (b) FTP (c)TCP (d) UDP

17. What is the process of converting a 3D scene into a 2D image called?
(a)Rasterization (b)Ray tracing (c) Shading (d) Modeling

18. What is the Internet of Things (IoT)?
(a) A network of interconnected devices (b) A cloud computing platform
(c)programming language (d) A type of database

19. OOP is stand for .
(a) Object-Oriented Programming (b0 Object-OF Programming
(c) Orient-object-programming (d) none

20.Which CSS property is used to set the background color of an element?
(a) color (b) background-color (c) font-size (d) margin

MATHEMATICS

21. Simplify: $(2x + 3y)^2$
(a) $4x^2 + 9y^2$ (b) $4x^2 + 12xy + 9y^2$ (c)$2x^2 + 6xy + 3y^2$ (d) $4x^2 + 6xy + 9y^2$

22. Solve the equation: $2x - 5 = 3x + 7$
(a) $x = -12$ (b)$x = 12$ (c) $x = 2$ (d) $x = -2$

23. Find the area of a triangle with base 8 cm and height 5 cm.
(a) 20 cm^2 (b) 16 cm^2 (c)40 cm^2 (d) 24 cm^2

24. The median of the data set {2, 4, 6, 8, 10} is:
(a) 2 (b) 6 (c) 8 (d) 10

25. Find the derivative of $f(x) = 3x^2 + 2x - 1$.
(a) $6x + 2$ (b) $6x^2 + 2x$ (c) $3x^2 + 2$ (d) $6x$

26. The integral of $2x\,dx$ is:
(a) $x^2 + C$ (b) $2x^2 + C$ (c) $x + C$ (d) $2x + C$

27. If A = [1 2; 3 4] and B = [5 6; 7 8], find A + B.
(a) [6 8; 10 12] (b) [6 10; 8 12] (c) [1 2; 3 4] (d) [5 6; 7 8]

28. A coin is tossed twice. What is the probability of getting two heads?
(a) 1/4 (b) 1/2 (c)3/4 (d) 1

29. Simplify: $\log_{10} 100$
(a) 1 (b) 2 (c) 10 (d) 100

30. Factorize: $x^2 - 9$
(a) $(x + 3)(x - 3)$ (b) $(x + 9)(x - 9)$ (c) $x(x - 9)$ (d) $x^2 - 9$

31. Simplify: $\log_2 8$
(a) 2 (b) 3 (c) 4 (d) 8

PHYSICS

32. A Cricketer hits 4 runs, when middle of the bat hits the ball. This is an example of Newton's ;
(a) 1st law of motion (b) 2nd law of motion (c) 3rd law of motion (d) law of gravity

33. The Distance between node and antinode is ;
(a) Lymda (b) lymda/2 (c) Lymda/4 (d) 2 Lymda

34. Mechanical Waves on the surface of liquid are ;
(a) Transverse (b) Longitudinal (c) Torsional (d) Both a and b

35.Torque acting on the bdy is given by ;
(a) r × F (b) rF (c) r× p (d) r × I

36. Inductor store energy in a ?
(a) Coil (b) Electric feild (c) Magnetic feild (d) None

37. The wavelength of light plays no rule in _____.
(a) Interference (b) Resolving power (c) diffraction (d) polarization

38.Which of the following is the SI unit of work ?
(a) Coulomb (b) Joule (c) Meter (d) Newton

39. The mass of the body can be converted into energy by expression _____.
(a) $E=mc^2$ (b) $E = m / c^2$ (c) $E=m^c$ (d) $E=m^2c$

40. Which of the following is not the effect of current ?
(a) Chemical effects (b) Compton's effect (c) Heating effect (d) Magnetic effect

41. Which of the following is the dimension of acceleration ?
(a) LT—1 (b) MLT—1 (c) LT—2 (d) LT^2

42. The area under a velocity-time graph represents.
(a) Acceleration (b) (b) Speed (c) Displacement (d) change in velocity

Quantitative Reasoning

Arithmetic (8mcqs)

43. Fractions: If 3/4 of a number is 12, what is the number?
A) 9 (B) 16 (C) 18 (D) 24

44. **Decimals**: What is the decimal equivalent of 3/5?
A) 0.35 (B) 0.6 (C) 0.53 (D) 0.75

45. **Percentages**: If 20% of a number is 50, what is the number?
A) 100 (B) 250 (C) 500 (D) 1000

46. **Ratio and Proportion**: If the ratio of boys to girls in a class is 3:2 and there are 15 boys, how many girls are there?
(A) 7 (B) 10 (C) 12 (D) 22

47. **Average**: The average of 5 numbers is 20. If 4 of the numbers are 15, 22, 25, and 18, what is the fifth number?

(A) 10 (B) 15 (C) 20 (D) 25

48.**Simple interest:** If $500 is invested at a simple interest rate of 50% per year, how much interest will be earned after 2 years ?
(A) $25 (B)$50 (C)$100 (D)$250

49. **Profit and loss :** If a shopkeeper buys an items for $100 and cells it for $120, what is his profit percentage?
(A) 10% (B) 15% (C) 20 (D) 25

Algebra and Function (06)

50.**Linear Equations:** Solve for x: 2x + 5 = 13
(A) 2 (B) 3 (C) 4 (D) 5

51. **Quadratic Equations:** Factor the quadratic: x^2 - 4
(A) (x + 2)(x + 2) (B) (x - 2)(x - 2) (C) (x + 2)(x - 2) (D) (x^2 + 4)

52.**Exponents:** Simplify: (2x^3)^2
(A) 4x^6 (B) 4x^9 (C) 8x^6 (D) 8x^9

53.**Radicals:** Simplify: √(16x^4)
(A) 4x^2 (B) 4x (C) 2x^2 (D) 2x

54.**Functions**: Evaluate f(3) for the function f(x) = 2x - 1
(A) 3 (B) 5 (C) 6 (D) 7

55. **Domain and Range:** What is the domain of the function f(x) = 1/(x-2)?
(A) All real numbers except 2 (B) All real numbers except -2
(C) All real numbers (D) All real numbers greater than 2

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Geometry (04 mcqs)

56. Angles: If two angles are supplementary, and one angle measures 75 degrees, what is the measure of the other angle?
(A) 15 degrees (B) 75 degrees (C) 105 degrees (D) 165 degrees

57.**Triangles:** In a right-angled triangle, if the hypotenuse is 10 cm and one leg is 6 cm, what is the length of the other leg?
(A) 4 cm (B) 8 cm (C) 12 cm (D) 16 cm

58.**Volume:** The volume of a cube with side length 4 cm is

(A) 16 cm^3 (B) 32 cm^3 (C) 64 cm^3 (D) 128 cm^3

59.**Polygons:** How many sides does a regular hexagon have?

(A) 4 (B) 5 (C) 6 (D) 7

Equations (04 mcqs)

60. Linear Equations: Solve for x: 3x + 7 = 16
(A) 3 (B) 4 (C) 5 (D) 6

61. **Quadratic Equations**: Factor the quadratic: x^2 - 9
(A) (x + 3)(x + 3) (B) (x - 3)(x - 3) (C) (x + 3)(x - 3) (D) (x^2 + 9)

62.**Exponential Equations**: Solve for x: 2^x = 32
(A) 3 (B) 4 (C) 5 (D) 6

63.**Logarithmic Equations**: Solve for x: log_3(x) = 2
(A) 6 (B) 7 (C) 8 (D) 9

Statistics (04 mcqs)

64.Measures of Central Tendency: The mean of the numbers 2, 4, 6, and 8 is
(A) 3 (B) 4 (C) 5 (D) 6

65. **Measures of Central Tendency**: The median of the numbers 2, 4, 6, and 8 is
(A) 3 (B) 4 (C) 5 (D) 6

66. **Measures of Central Tendency:** The mode of the numbers 2, 2, 4, 6, and 8 is
(A) 2 (B) 4 (C) 6 (D) 8

67.**Measures of Dispersion:** The range of the numbers 2, 4, 6, and 8 is
(A) 2 (B) 4 (C) 6 (D) 8

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Mental Mathematics (06 Mcqs)

68. Square Roots: What is the square root of 144?
(A) 10 (B) 11 (C) 12 (D) 13

69. **Exponents:** Evaluate 2^3.
(A) 6 (B) 7 (C) 8 (D) 9

70. **Mental Multiplication:** Multiply 12 by 8.
(A) 86 (B) 96 (C) 106 (D) 116

71.**Algebra:** Simplify: 3x + 2x - 5x
(A) 0 (B) 1 (C) 2 (D) 3

72.**Fractions**: What is the value of 3/4 + 1/2?
(A) 1 (B) 1 1/4 (C) 1 1/2 (D) 2

73. **Percentages:** Calculate 20% of 150.
(A) 20 (B) 30 (C) 40 (D) 50

Verbal Reasoning

Choose the lettered words or phrase that is most NEARLY OPPOSITE in meaning to the word in capital letters.

74. We are going to **WIND-UP** our business which our grandfather had _____ in 1947.
(a) Closed (b) decorated (c) designed (d) established

75. I had to _____ all my shares immediately after I **BOUGHT** them .
(a) decorate (b) distribute (c) purchase (d) sell

76. Some people have **OPTIMISTICS** approaches towards life rather than a _____ one.
(a) logistic (b) majestic (c) nostalgic (d) pessimistic

Four lettered pairs (A to D) follow a related pair of words given in the questions. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair in capital letters.

77. BOAT: OAR ::
(a) book : page (b) car :steering wheel (c) teacher : pupils (d) sweet: sour

78. SIGULAR : PLURAL ::
(a) below : beneath (b) haughty : vain (c) least : most (d) lift : soar

79. OCEAN : WEEDS ::
(a) kids : parent (b) movie : actors (C) loyal : state (d) forest : bushes

80. FINGER : GUITAR ::
(a) beak : birds (b) level: vertical (c) snake : mice (d) stick : drum

81. TEETH : BITE ::
(a) knife : cut (b) tree : green (C) drop : ocean (d) language : alphabet

82. SUMMARY : PRECIS ::
(a) ladle : pan (b) monitor : Screen (c) order : obedience (d) reverse : push

Select the Similar (Synonyms)

83. Dichotomy
(a) Contrast (b) Willingness (c) Inclination (d) Disposition

84. Pragmatic :
(a) idealistic (b) Realistic (c) Irrational (d) Illogical

85. Real :
(a) Copied (b) idea (c) original (d) Given

86. Absolute :
(a) Division (b) Complete (c) Small (d) Half

Pick the correct one choice (TENCES)

87. He invited _____ and Nasir to dinner.
(a) we (b) you (c) us (d) them

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88. Neither they nor their friend _____ solved the question.
(a) has (b) have (c) was (d) were

89. He asked me _____?
(a) how are you (b) how was I? (c) how are you ? (d) How I was.

90. She asked me _____?
(a) bring a glass of water for her. (b) to bring a glass of water for me .
(c) to bring a glass of water for her. (d) none of these

91. When your friends _____ , he'll be very tired .
(a) arrived (b) will arrive (c) arrives (d) is arriving

92. The sentence which has one dependent and one independent class is called _____ .
(a) Compound Sentences (b) Complex sentences (c) Compound Complex Sentence (d) none

93. If I _____ a million rupees, I would help the poor.
(a) has (b) have (c) had (d) should has

Comprehensions

The world is becoming increasingly interconnected. People from different cultures are interacting with each other more frequently than ever before. This globalization has led to both positive and negative consequences. On the one hand, it has promoted economic growth and cultural exchange. On the other hand, it has also contributed to environmental degradation and social inequality.

94.According to the passage, what is the main consequence of globalization?

- A. Increased cultural diversity
- B. Economic stagnation
- C. Environmental protection
- D. Social equality

95.What is the negative aspect of globalization mentioned in the passage?

a. A. Economic growth
b. B. Cultural exchange
c. C. Environmental degradation
d. D. Social equality

Passage 2:

Reading is a fundamental skill that is essential for success in education and beyond. It is not only about recognizing words and understanding their meanings, but also about interpreting the text and making connections between different ideas. Effective reading involves active engagement with the material and critical thinking.

96.What is the primary purpose of reading?

e. A. To memorize words
f. B. To understand ideas
g. C. To avoid critical thinking
h. D. To become a passive learner

97.What does effective reading require?

i. A. Passive engagement
j. B. Critical thinking
k. C. Memorization of words
l. D. Avoidance of connections

Passage 3:

Technology has transformed the way we live and work. It has made communication easier, access to information faster, and entertainment more convenient. However, excessive use of technology can also have negative consequences, such as addiction, social isolation, and health problems. It is important to use technology wisely and balance its benefits with its potential drawbacks.

98.What is the main benefit of technology?

m. A. Addiction
n. B. Social isolation
o. C. Health problems
p. D. Improved communication

99.What is the negative consequence of excessive technology use?

q. A. Improved communication
r. B. Addiction
s. C. Easier access to information
t. D. More convenient entertainment

Passage 4:

Climate change is a pressing global issue. It is caused by human activities, such as burning fossil fuels and deforestation. The consequences of climate change include rising sea levels, extreme weather events, and biodiversity loss. Addressing climate change requires international cooperation and sustainable practices.

100What is the main cause of climate change?

u. A. Natural phenomena
v. B. Human activities
w. C. Sustainable practices
x. D. International cooperation

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THE END

